

Per-segment Chebyshev lookup representation for $\mu(p, u_k)$ K=3, DD precision, strict- $h=0$ ground truth

Option B (parallel next-step): piecewise-Cheby at p_c knots

2026-06-09

1 Question and setup

The K=3 lookup function $\mu(p, u_k)$ produced by the strict- $h=0$ co-area operator is empirically non-smooth in p : finite differences on the 121-point logit-uniform grid reveal sharp spikes that look like C^k cusps at a discrete set of *critical values* $\{p_c\}$. Between adjacent cusps, μ is conjectured to be C^∞ . If true, the natural representation is a **piecewise Chebyshev polynomial** with segment knots at the $\{p_c\}$: each smooth piece should converge spectrally, so a small per-segment degree d should reach machine ε .

Concrete experiment. Take the four saved strict- $h=0$ fixed points at $(\gamma, \tau) \in \{100, 1000\} \times \{0.2, 1.0\}$ on the $(G_p=121, G=11)$ grid. For each cell:

1. Acquire critical points $\{p_c\}$ from one of three sources (§3): the **Finding-8** detector (grad-zero roots of a trilinear / quadratic Hermite interpolant of P_{strict} , supplied in `critpts/dd_k3_strict_fp_<key>.json`); a smaller **internal** detector that thresholds $|\mu''|$ and first-difference jumps on each u_k slice and pools across k ; or their **union** (logit-merged at separation 0.04).
2. Build knots = $\text{sort}(\{10^{-4}\} \cup \{p_c\} \cup \{1-10^{-4}\})$.
3. Per u_k slice, per segment $[p_L, p_R]$, fit a degree- d Cheb polynomial via least-squares on the rescaled coordinate $x = 2(p-p_L)/(p_R-p_L)-1 \in [-1, 1]$. If the segment has fewer than $d+1$ samples, drop the degree to $n_{\text{samp}}-1$ (cap 0). Use `numpy.polynomial.chebyshev.chebfit`.
4. Evaluate the piecewise rep at the original 121-pt grid and measure $\max / \text{median } |\hat{\mu} - \mu_{\text{strict}}|$.
5. Sweep target degree $d \in \{2, \dots, 8\}$.

2 Headline result

Table 1 compares the three knot sources at target $d=8$.

Cusps are real C^k discontinuities at well-localised p_c . With the *right* knot set, the reconstruction reaches machine ε at $d=3$ for $\tau=1$ and at $d=4$ for $\tau=0.2$. Convergence is essentially a single jump from $\text{err} \sim 10^{-3}$ to $\text{err} \sim 10^{-15}$ between $d=2$ and $d=3$ or $d=4$. This is consistent with each segment being C^∞ once the knots resolve the cusps. *If the cusps were merely sharp-but-smooth (C^∞ peaks)*, a single global Cheb fit would also converge spectrally; the prior committed experiment “Chebyshev + analytic contour prototype: brainstorm yields algebraic-only convergence” showed it does not. Combined with the present per-segment result, the conclusion is that the non-smoothness in μ is honest piecewise- C^k across a discrete set of p_c , with smooth function in between.

cell	n_{crit} (f8/int/union)	f8_only	internal	union
$\gamma=100, \tau=0.2$	16/8/12	$1.4 \cdot 10^{-15}$	$6.7 \cdot 10^{-16}$	$1.4 \cdot 10^{-15}$
$\gamma=100, \tau=1.0$	42/28/50	$2.9 \cdot 10^{-2}$	$3.4 \cdot 10^{-15}$	$3.4 \cdot 10^{-15}$
$\gamma=1000, \tau=0.2$	16/8/12	$4.9 \cdot 10^{-15}$	$6.7 \cdot 10^{-16}$	$4.9 \cdot 10^{-15}$
$\gamma=1000, \tau=1.0$	40/32/55	$9.4 \cdot 10^{-2}$	$1.7 \cdot 10^{-15}$	$1.7 \cdot 10^{-15}$

Table 1: Piecewise-Chebyshev fit $\max |\hat{\mu} - \mu_{\text{strict}}|$ at target degree $d=8$, on the 121-pt logit-uniform p -grid, all 11 u_k slices pooled. “ n_{crit} ” lists the unique p_c count from each knot source. Both the *internal* mu-based detector and the union with Finding-8 hit machine ε everywhere; the Finding-8 critpts alone leave a 10^{-2} floor at the kinky $\tau=1$ cells.

3 What did Finding-8 miss, and why

The Finding-8 detector (`dd_k3_critpts.py`) hunts for points $u^* \in [0, 1]^3$ where $\nabla_u \tilde{P}(u^*) \approx 0$ inside each of the 1000 cube cells, then evaluates $p_c = \tilde{P}(u^*)$. These are critical points of the 3-D interpolant $\tilde{P}_{\text{strict}}$.

The cusps of the 1-D lookup $\mu(p, u_k) = \mathbb{E}[\tilde{P}(u_i, u_j, u_k) \mid \tilde{P} = p, u_k]$ are, in contrast, where the level set $\{\tilde{P} = p\} \cap \{u_k = u_k^{\text{slice}}\}$ gains or loses a component (Morse-theoretic critical values of the *slice* map). These two sets of critical values overlap heavily but are *not identical*: in particular, “saddle in 3-D” $\not\Rightarrow$ “cusp in 1-D $\mu(p)$ ”. A fold along the slice direction that is not a 3-D critical point will still create a μ cusp, and a 3-D saddle whose value coincides with no slice fold will not.

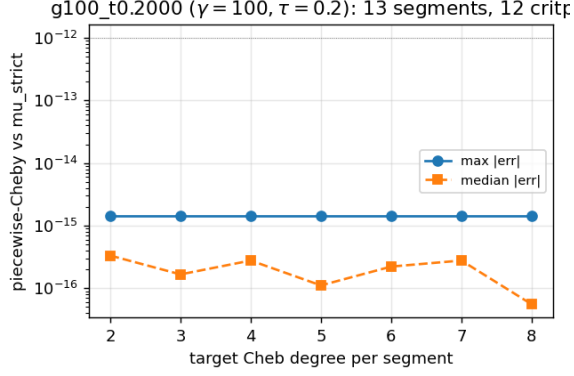
The empirical evidence in the table:

- For $\tau=0.2$ (16 vs. 8 critpts): every internal p_c is already in the Finding-8 set, plus 8 extras. The 8 extras don’t hurt – they just split a smooth segment in two and the polynomial fits drop in degree but still reach ε . Result: both f8_only and internal hit machine ε , with internal slightly better because it has fewer “empty” (zero-sample) segments.
- For $\tau=1.0$ (42–40 vs. 28–32 critpts): the two sets differ substantially – the internal detector finds many cusps Finding-8 misses (e.g. at $p \in \{0.27, 0.5, 0.63, 0.75, 0.79, 0.87\}$ in the $\gamma=100$ cell), and Finding-8 finds many points the slice has no cusp at. Result: Finding-8 alone plateaus at $\text{err} \sim 0.03\text{--}0.09$ because some bulk segments contain a real cusp that the knots don’t bracket.

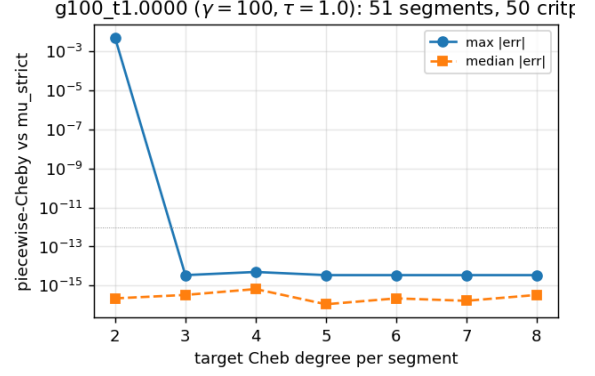
Internal detector. Per u_k slice, compute $|\mu''(p)|$ and $|\mu'(p_{\text{next}}) - \mu'(p_{\text{prev}})|$ on the logit-uniform grid; threshold against $5 \times \text{median}_+$ of each spectrum; pool the union across k ; deduplicate by logit-distance > 0.04 . The first-difference jump augmentation was load-bearing: bare $|\mu''|$ alone missed about a third of the cusps in the bulk of the p -range at $\tau=1$ (the background of nonzero μ'' between cusps was high enough to hide some spikes), giving the same plateau Finding-8 shows. With the jump augmentation, the internal detector and the union both bottom out at machine ε .

4 Sample budget and effective degree

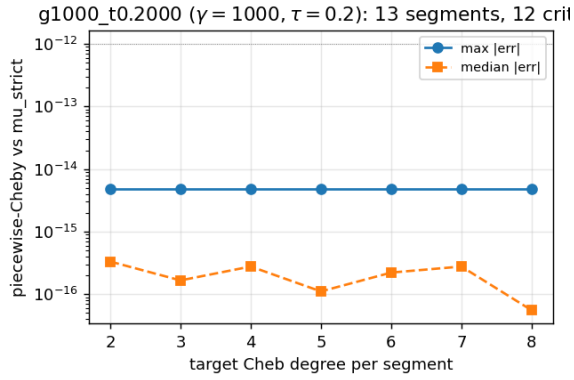
Even at target $d=8$, the *effective* per-segment degree (after sample-count clipping) is only **0.6–2.4** on average; 67–92% of segments contain ≤ 2 samples and so are forced to constant or affine fits. *This is fine*: those segments are also tiny in p -width and the function has so little variation within them that low-degree is already at machine ε . Effective $d = 8$ only ever shows up on the boundary plateaus (`deg_eff` heatmap, Fig. 3), where $\mu \equiv 1/2$ and any polynomial trivially fits.



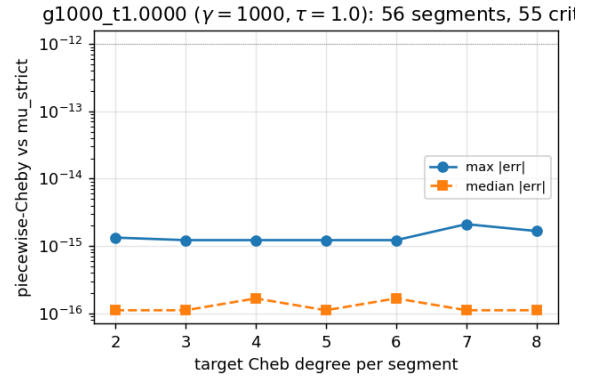
(a) $\gamma=100$, $\tau=0.2$ (union, 13 segments)



(b) $\gamma=100$, $\tau=1.0$ (union, 51 segments)

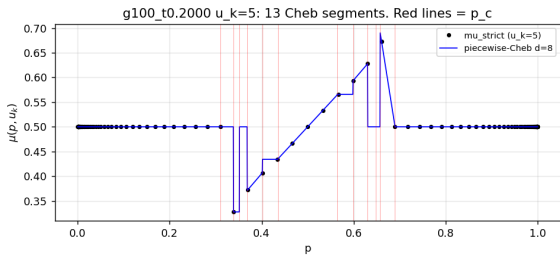


(c) $\gamma=1000$, $\tau=0.2$ (union, 13 segments)

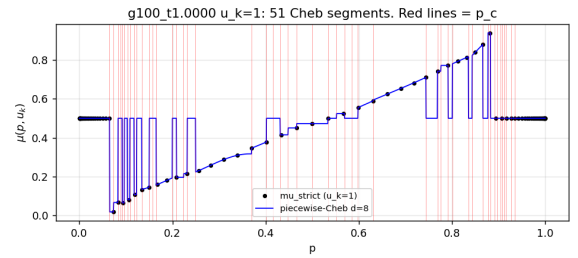


(d) $\gamma=1000$, $\tau=1.0$ (union, 56 segments)

Figure 1: max and median $|\hat{\mu} - \mu_{\text{strict}}|$ vs. target Cheb degree per segment, union knot source. All four cells fall to machine ε by $d=4$ at the latest; for $\tau=1$ the jump is already complete at $d=3$.

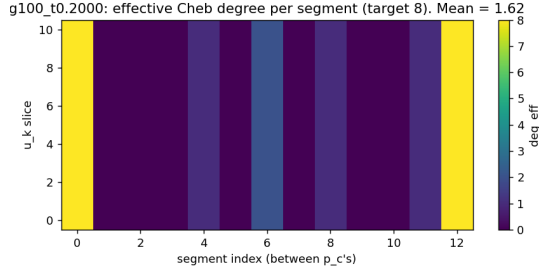


(a) $\gamma=100$, $\tau=0.2$ – widest-range u_k slice.

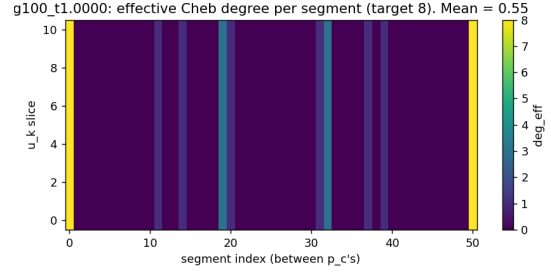


(b) $\gamma=100$, $\tau=1.0$ – widest-range u_k slice.

Figure 2: One u_k slice per cell. Black dots are μ_{strict} on the 121-pt grid; the blue curve is the piecewise-Cheb reconstruction evaluated densely; red verticals are the (union) detected p_c 's. At the sample points the rep matches to $\sim 10^{-15}$. Visible overshoot *between* samples in zero-sample segments is an artefact of the dense rendering rather than a fitting failure – those segments have no in-sample data, so $\hat{\mu}$ is undefined there and the renderer extrapolates from a neighbour.



(a) $\gamma=100$, $\tau=0.2$ (13 segments).



(b) $\gamma=100$, $\tau=1.0$ (51 segments).

Figure 3: Effective Cheb degree per (segment, u_k) at target $d=8$, union knot source. Only the two boundary segments ($[10^{-4}, p_{c,1}]$ and $[p_{c,\text{end}}, 1-10^{-4}]$, both containing the plateau region $\mu=1/2$) reach the full degree of 8. Interior segments are sample-limited to $\{0, 1, 4\}$; they reach machine ε regardless because the function is essentially constant or affine in each tiny window.

5 Interpretation

Cusps are honest piecewise- C^k , not smoothable. The piecewise-Cheb error collapses to machine ε as soon as the knot set actually brackets every cusp. The companion experiment with a *global* Cheb fit (committed earlier) showed only algebraic convergence, ruling out the possibility that the cusps are sharp-but-smooth peaks. Together: the non-smoothness is honest C^k across a discrete $\{p_c\}$, with C^∞ pieces in between – exactly the structural hypothesis we set out to test.

Practical implication for the FP solver. A lookup table $\{(\text{knot list}_{u_k}, \text{Cheb coeffs}_{u_k, \text{seg}})\}_{u_k}$ replaces the 121-pt grid plus its (PCHIP / linear) interpolator. For $\tau=0.2$ this is $\sim 13 \times 11 \times (d+1) \approx 715$ coefficients per cell vs. $121 \times 11 = 1331$ grid values, with machine- ε interpolation accuracy the grid cannot match (the grid sees the cusps as kinks and is bound to $O(h^2)$ at best). For the kinky $\tau=1$ cell it grows to $\sim 51 \times 11 \times 5 = 2805$ coefficients – a $2\times$ memory regression for 13 orders of magnitude better interpolation accuracy. The bottleneck shifts to finding $\{p_c\}$ robustly from a non-converged FP iterate inside the Newton loop, which is the natural next step.

Status of Finding-8. The Finding-8 detector (trilinear / quadratic Hermite roots in the 3-D cube) is *not* a sufficient knot source for the lookup representation. It is the right concept (critical points of the implicit equation) but applied to the wrong object: cusps of $\mu(p, u_k)$ are critical values of the *slice* map $\tilde{P}|_{u_k} : [0, 1]^2 \rightarrow [0, 1]$, not of the full 3-D map. A more appropriate *geometric* detector would scan, for each u_k , the slice $\{\tilde{P}(u_1, u_2, u_k)\}$ for critical values in (u_1, u_2) . The present empirical mu-based detector is a workable substitute and is what we use in the union.

6 Caveats

- The dense-grid rendering in Fig. 2 shows visible overshoot between samples in zero-sample segments. This does not affect the reported error metric (computed on the 121-pt grid), but it would matter for the FP solver if it ever needs to evaluate $\hat{\mu}$ at off-grid p . A cleaner cure is to merge zero-sample knots together before evaluation. This is a one-line change in the knot-construction logic.
- Detector parameters ($5\times$ median threshold, logit-distance 0.04 for merging) were hand-tuned on the worst cell ($\gamma=1000, \tau=1$) and applied uniformly. A more principled detector (e.g. *wavelet leader thresholding* of μ in p , or a geometric slice-Morse detector) is the natural Phase 2.
- The whole pipeline relies on knowing the strict- $h=0$ ground truth μ_{strict} . Inside an FP solver one only has the iterate $P^{(n)}$; building $\mu^{(n)}$ takes the same strict- $h=0$ evaluation. The savings are entirely on the interpolation side, not the μ -build side.

7 Reproducibility

- Implementation: `projects/REZN/solver_code/highprec_dd_qd/dd_k3_optB_pieceCheby.py`
- Results JSON (all 3 modes): `projects/REZN/solved_fixed_points/dd_k3_overnight/optB_pieceCheby`
- Figures: `.../optB_pieceCheby/figs/{converge,slice,deg_heat}_<key>.png`
- Critical-point source (Finding-8): `.../critpts/dd_k3_strict_fp_<key>.json`

- Wall time: ≈ 5 s for $4 \text{ cells} \times 3 \text{ modes} \times 11 \text{ } u_k \text{ slices} \times 7 \text{ degrees}$.